Jackson Adams

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EDUCATION

Miami University Oxford, Ohio
Bachelor of Science Expected Graduation: May 2029
Major: Computer Engineering
Minor: Deep Learning and Artificial Intelligence
GPA: 4.00, President's List Fall & Spring 2025

- Earned the Redhawk Excellence Scholarship
- **Organizations:** Honors College, Theta Tau Engineering Fraternity, Miami Tough Mudder's

PROJECTS

Simulated Pick-and-Place Automation Cell (PLC & HMI)

- Developed a PLC control program in CODESYS ladder logic for a simulated conveyor, pneumatic clamp, and robot pick-and-place cell.
- Implemented a state machine sequencer with sensor-confirmed step transitions & PLC-to-robot handshake.
- Designed timeout-based fault detection with latched alarms, E-stop interlocks, and an operator HMI for control, status, and fault messages.

AI Trading Card Scanner & Portfolio App (Foilio) https://github.com/jgadams15/Foilio

- Built an on-device computer vision pipeline using PyTorch / Meta's DINOv2 to detect and identify cards from the camera feed in real time, with no server calls or scan limits.
- Integrated the TCGdex API and eBay/TCGplayer market data to show live card prices across 160 sets / 20,000+ cards
- Used Claude Code for AI-assisted development while owning architecture, code review, debugging, and testing

Lego NXT Robotics Control Systems

- Developed a multi-tasking control system for a LEGO NXT robot using NXC and rotation-sensor feedback.
- Implemented mutex-based task coordination to safely manage simultaneous movement and sensor-monitoring routines.
- Integrated touch-sensor logic to trigger dynamic behavior changes and interrupt movement tasks.

WORK EXPERIENCE

Undergraduate Teaching Assistant: Fundamentals of Programming August 2026 - Present

Miami University Oxford, Ohio

- Assisted a class of 30 students on the introduction of basic coding mechanics in python. Graded and reviewed student work.
- Guided students through debugging by analyzing their code, identifying logic errors, and explaining underlying programming concepts in real time.

Host May 2023 - July 2025

Carrabba's Italian Grill Canton, Ohio

- Greeted and seated 75-150 guests and managed 40-60 reservations a shift ensuring a smooth start to dining experience
- Trained and onboarded 5 new hosts, providing hands-on coaching in guest service, reservations, and front-of-house operations to ensure consistent, high-quality performance

SKILLS

- Programming: Python, Java, HTML, CSS, Git/GitHub
- Technical Skills: PLC Programming (Ladder Logic), FPGA Design, MATLAB, Arduino, Excel